De(ep)Composition: Deep Research for Argumentation by Query Decomposition*

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Abstract

Searching the web in an iterative process to answer a question and produce a cited report, leading systems such as OpenAI Deep Research and STORM summarize the evidence for each side but do not take a position. We present De(ep)Composition, a deep research system designed to generate arguments, which decomposes a question into a graph of sub-questions, answering each narrow query on the web, and combining the answers for each sub-question into an argumentative report that takes a position. In blind evaluation on 10 contentious questions, De(ep)Composition takes a firm position in 9 reports, compared to 1 for OpenAI Deep Research and 2 for STORM. In six automated debates against OpenAI Deep Research, it holds its assigned side in 91% of turns (vs. 68%).

De(ep)Composition cites a higher fraction of claims (0.80, vs. 0.68 and 0.60), uses a larger share of its cited sources (0.76, vs. 0.67 and 0.61), achieves the highest logical coherence score (7.7 out of 10, vs. 6.5 and 6.0) and uses fewer hedging words than STORM (2.8 per 1k words vs. 6.4) and more commitment words (6.8 vs. 3.8). Its reports are shorter than STORM's (median 1339 words vs. 2426) and cite more densely (5.7 vs. 4.7 citations per 100 words), but draw on fewer distinct sources (8 vs. 27). An automatic check of the fidelity of citations (which picks out the correct source for 96% of known-correct claims) finds that in our 10 evaluated reports, cited sources are the best match for only 14%, near the 10% expected by chance. We trace the likely cause to how citation numbers are handled when answers are combined, and giving each source a single global number restores the best match to 96% in a replay of a logged run.

1 Introduction

Deep research systems automate answering questions by asking multiple rounds of search queries. Each round involves reading search results and deciding if more queries are needed. The system eventually writes a report citing the page in which each fact appears. Two of the most well-known systems are OpenAI Deep Research (OpenAI, 2025), released in February 2025, and STORM (Shao et al., 2024), from Stanford OVAL.

OpenAI Deep Research and STORM both produce summaries of information rather than taking a position on the information they encountered. For example, on the question "Should the Federal Reserve keep interest rates high?", the output is a set of evidence from which a reader can draw their own conclusions. While this is great for a starting point, many professional contexts require a final conclusion. A lawyer writing a brief makes a case for their client's side, and an analyst provides a recommendation to buy or sell an investment. In such contexts, the reader expects the writer to take a side and to provide claims with supporting evidence.

The distinction between a survey and an argument affects how research should be structured. A survey can be built incrementally, with each search round informing the next. The

*Code and data: https://github.com/jessiedong01/deep-research-for-argumentation

content of the report is ultimately decided by the search results. In contrast, an argument has a defined structure before searching. A claim has a few sub-claims, each of which must be supported by retrievable evidence. Research to generate an argument can be planned in advance by breaking down the argument into narrow sub-questions, which can be answered with search and combined together in argument order.

Our system, which we abbreviate De(ep)Composition (combining "deep research" and "decomposition"), uses the following pipeline. When given a question, a language model produces a directed acyclic graph of sub-questions, with instructions on how to combine the answers of child sub-questions into the parent node's. The leaf nodes are solved with retrieval-augmented generation, and the parent nodes are composed from their children in topological order. If the composed answer is incomplete, the system adds a new subtree of sub-questions below. The final graph is converted to an outline and to the actual report arguing for the root node's answer. Figure 1 illustrates.

We evaluate this pipeline against OpenAI Deep Research and STORM, showing results from six automated debates and blind rater evaluation of 10 reports. Our pipeline receives the highest scores for argument quality and takes a side far more often than the baselines. A secondary analysis of the reports we retained, which does not use a rater, supports the finding that our pipeline takes a side more often, but qualifies the other two results. The reports have dense citations but few unique sources, the conclusions are often more modest than the titles, and an automated test for attribution indicates that when a fact is cited, the citation rarely points to the claim's original source. This problem is caused by a flaw in how our pipeline manages citations during composition, a flaw that can affect any system that stages text composition with citations.

The rest of the paper is organized as follows. §3 defines the task and highlights the challenge of retaining citations after composition; §4 describes the pipeline; §5 and §6 present the initial evaluation and its limitations; §7 analyzes the contents of the reports; §8 describes an attribution test and shows how it identifies the flaw in our pipeline; §9 explains what the results are and are not evidence for.

2 Related Work

OpenAI Deep Research (OpenAI, 2025) is a search planning and browsing agent that produces reports with citations. It is trained with reinforcement learning on browsing tasks of the kind collected in BrowseComp (Wei et al., 2025), in which the response is a short fact that is challenging to find. The goal is to find and cite relevant information, rather than to argue a position. Its intermediate reasoning remains within one context window, without explicit dependencies between sub-questions.

STORM (Shao et al., 2024) generates Wikipedia-like articles. It first discovers different perspectives on a topic, then simulates conversations in which the writers of those perspectives query an expert with search results, then structures the answers into an outline and article. Its sub-questions are generated via discussions throughout research (bottom-up), and its output is intentionally balanced. Since they target surveys, we view them as the strongest summary-oriented baselines rather than systems for argumentation.

Decomposing a problem into sub-problems is a well-studied prompting technique. In least-to-most prompting (Zhou et al., 2023), decomposed prompting (Khot et al., 2023) and self-ask (Press et al., 2023), the model generates sub-questions that, when answered, together provide a solution to the original question. Tree of Thoughts (Yao et al., 2023a) conducts a search over intermediate steps, ReAct (Yao et al., 2023b) interleaves reasoning with tool calls, and Graph of Thoughts (Besta et al., 2024) generalizes the tree to a graph where intermediate results can merge. These methods are designed for problems with an objective, verifiable answer (e.g., arithmetic, puzzles, short-form Q&A) and do not generate prose from retrieved evidence, or revise their structure when a combined answer proves incomplete. De(ep)Composition uses decomposition, but extends these ideas by explicitly instructing a model how to compose answers, providing answers conditioned on retrieved evidence, and refining the graph when gaps in the solution are identified.

AutoGen (Wu et al., 2023), CAMEL (Li et al., 2023), and natural-language societies of mind (Zhuge et al., 2023) use multiple language-model agents conversing on roles such as planner, researcher, and critic. The agents pass back and forth messages, but the task structure is not represented, so one agent's output does not formally constrain how another uses it in the final composition. In our design, the argument structure is represented explicitly, via a DAG, so that answers to sub-questions formally constrain how the parent's answer is composed.

Argument mining (Lawrence & Reed, 2020) extracts claims and premises from text and their logical relations. Argument generation generates arguments from scratch. Hua & Wang (2018) introduce a seq2seq model conditioned on evidence to generate counter-arguments. Project Debater (Slonim et al., 2021) extracts claims and evidence from a large mined corpus to generate debate speeches. In these methods, the corpora of claims and evidence is fixed, and text generation is done in a single step. In contrast, De(ep)Composition searches the web for evidence and then composes the argument structure before filling it in.

The literature contains studies of how each citation supports the sentence containing its in-text anchor. This is called attribution (Rashkin et al., 2023, Liu et al., 2023). Liu et al. (2023) find, for four generative search engines, that about half of the sentences in the generated text are fully supported, and about three-quarters of the citations support their containing sentence. In another study, Gao et al. (2023) create a citation benchmark using an entailment model to automatically calculate citation precision and recall. As these studies are on the task of generation conditioned on retrieved passages in a single step, they do not address the more challenging multi-step setting we consider, in which citations must be correct over multiple rounds of composition. We perform a simple lexical test, requiring no additional model, to check whether citations are retained in the composed answer.

We use DSPy (Khattab et al., 2024) to implement our pipeline. DSPy allows one to write code that interprets each language-model call as a signature, consisting of an input, an output, and an instruction for the model to follow. These calls can then be composed into a program. All signatures are listed in Appendix A.

3 Problem Formulation

Problem: Given a motion m (e.g., "The U.S. Federal Reserve should prioritize maintaining high interest rates over stimulating growth"), the goal is to generate an argumentative report R, which takes a position on the motion, argues for its position with claims, and anchors each factual claim to its source retrieved from the web. We define a good report as one that is (i) *decisive* (takes a single position and argues for it), (ii) *grounded* (each claim cited is supported by its source, known as verifiability (Rashkin et al., 2023, Liu et al., 2023)), and (iii) *connected* (the claims make a coherent case rather than a list of facts).

De(ep)Composition performs its research by constructing a directed acyclic graph $G = (V, E)$, with a single root node r, whose question is m. Each node $v \in V$ has a question q_v, a format for the expected answer f_v, an answer a_v, and a set of sources S_v. There is an edge $(u, v) \in E$ if the question q_v is a sub-question of the question q_u. For leaf nodes $v \in V$, the answer is obtained by retrieval, $a_v = \textsc{Answer}(q_v, S_v)$, with S_v being the documents we retrieved. For a parent node $u \in V$ with children $C(u)$, the answer a_u is obtained by composition,

$$a_u = \textsc{Compose}\big(q_u,\ \{a_v : v \in C(u)\},\ \kappa_u\big), \qquad S_u = \textstyle\bigcup_{v \in C(u)} S_v, \tag{1}$$

where κ_u are the composition instructions for how to combine its children's answers. The report is written from the answered graph, $R = \textsc{Write}(m, G)$.

A report can be grounded only if its citations stay correct through every composition step. The provenance of a citation is considered valid only if all citations made along the way are correct. If $\text{cite}(a_v)$ indicates the set of (claim, source) pairs in a_v, then composition preserves provenance if every (claim, source) pair that a_u borrows from a child is one the child itself contains. Thus, any citation can be traced through the graph all the way to the

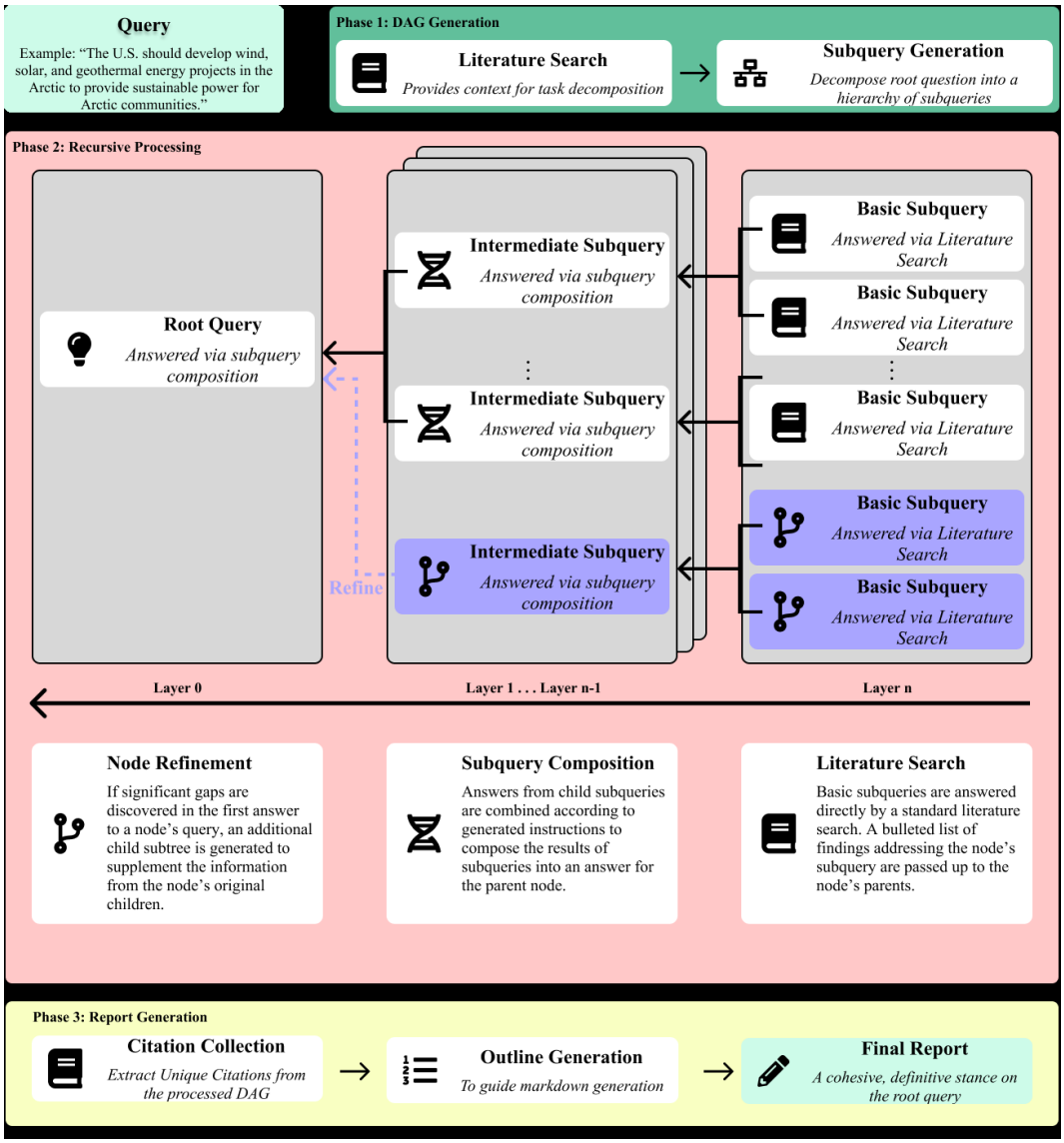


Figure 1: The three phases of De(ep)Composition. In Phase 1, a literature search is used to contextualize the question and decompose it into a DAG of sub-questions. In Phase 2, leaves are answered using literature search, parents are composed from their children, and then the pipeline iterates to refine nodes with incomplete answers by adding new sub-trees. In Phase 3, the fully completed DAG is converted into an outline and a report that justifies the root answer.

original source leaf. Section 8 will quantitatively assess the provenance preservation in the pipeline, and pinpoint where failures occur.

4 Method

De(ep)Composition takes a research question as input and returns a report that is properly citation-backed and advances an answer to the question. It works in three phases: DAG creation, recursive answering, and report crafting. This section describes the pipeline version used to generate the reports in Section 5; code versions are listed in Appendix C. All

calls to language models in the pipeline are made using DSPy signatures (Khattab et al., 2024) (details in Appendix A).

Literature search. The planning phase and leaf nodes both use the same literature search process. First, a planning model generates web search queries from the question. A retriever uses the Serper API to fetch results and extract page content, and then uses `text-embedding-3-large` to embed all passages and select the most relevant ones for the query. An answer model then answers the query with citations from the passages. After each iteration, a completion model decides if another round of search is necessary, limited by a maximum number of retriever calls per question.

Phase 1: DAG generation. To begin, a brief literature search is done on the main question to provide context. Then, the planning model creates the DAG layer-by-layer, asking child sub-questions from every node in the current layer. For each node, the model uses the node's question, the literature search summary, the children nodes created so far, and the entire graph structure to create the children. Using the entire graph prevents duplicate sibling questions and encourages the children of a node to jointly address the question. There are limits placed on the depth of the graph, the number of nodes, and the number of children per node.

Another pass generates composition instructions for every parent. These instructions are based on the parent's question, the parent's desired output format, and the children's questions. The model is asked to explain how the children's answers can be combined to generate an answer for the parent. The instructions use unique identifiers for children to make all the dependencies clear.

Phase 2: recursive answering. The DAG is topologically sorted, and then processed bottom-up, since parents are composed from answered children. Leaves are answered in parallel using literature search, with the answer model generating bullet points using the desired format while preserving all citations. Then parents are composed in topological order. The synthesis model uses the children's answers, the composition instructions, and the desired output format to generate the parent's answer. The model is asked to use only text from the children's answers and use the same citation numbers from the children's answers. Additionally, the model must indicate any contradictory claims between children. Since parents only depend on children, any citation made in the parent's answer should trace back to evidence gathered at a leaf.

Gap exploration and refinement. It is possible that the initial plan will miss something that the research later shows to matter. Thus, after composing a parent, a gap explorer identifies any potential gaps in the answer. A gap evaluator then evaluates if the gap is important, at a configurable conservatism level. If it is, a new sub-tree is generated of questions addressing the gap, and then attached under the parent. The new sub-tree is processed in the same way as the rest of the graph, and the parent is re-composed. In our current setup, a parent will only see one pass of this refinement.

Phase 3: report generation. After the DAG is completed, it is converted to text, and all citations are unified to a single list of numbered citations. Next, the report is outlined and generated. An outline model is asked to advocate for a position on the root question, and then generates an outline where each section maps to each branch in the graph. Then, the report is generated using the outline and the graph. The model is asked to use the same citations, advocate for the root answer's claim, and name the report the claim rather than the question. Finally, the pipeline standardizes the numbering and adds a bibliography.

Let $|V_{\text{leaf}}|$ be the number of leaves in the finalized DAG, and let $|V_{\text{par}}|$ be the number of parent nodes. Each leaf node makes 1 literature search call, which includes generating a search query, retrieving with embedding, making an answer call, and checking completeness. Each parent node makes 1 composition call per refinement round, and 1 gap check. For planning, each parent node makes 1 planning call to plan the children, and 1 planning call for instructions. Additionally, writing has 2 calls. Overall, the total approximate num-

Algorithm 1 De(ep)Composition

Require: motion m, limits on depth d, nodes n, and children c, refinement rounds ρ

$L \leftarrow \textsc{LiteratureSearch}(m)$ ▷ context for planning
$G \leftarrow$ root r with $q_r = m$
for each layer $\ell = 0, \ldots, d-1$ **do** ▷ Phase 1
  for each node u at depth ℓ **do**
    add children $\textsc{PlanChildren}(q_u, L, G, c)$ under u, within the node limit n
  end for
end for
for each parent u **do** $\kappa_u \leftarrow \textsc{CompositionInstructions}(q_u, \{q_v : v \in C(u)\})$
end for
for each node u in topological order, leaves first **do** ▷ Phase 2
  if u is a leaf **then**
    $(a_u, S_u) \leftarrow \textsc{AnswerFromSearch}(q_u)$
  else
    for $k = 0, \ldots, \rho$ **do**
      $a_u \leftarrow \textsc{Compose}(q_u, \{a_v\}_{v \in C(u)}, \kappa_u), \quad S_u \leftarrow \bigcup_{v \in C(u)} S_v$
      $\Delta \leftarrow \textsc{FindGaps}(q_u, a_u)$
      if $k = \rho$ **or** $\Delta = \emptyset$ **then break**
      end if
      attach a sub-tree for Δ under u and process it as in Phase 2
    end for
  end if
end for
$O \leftarrow \textsc{Outline}(m, a_r, G), \quad R \leftarrow \textsc{Report}(O, m, a_r, G)$ ▷ Phase 3
return R with a bibliography of $\bigcup_v S_v$

ber of model calls is $4|V_{\text{leaf}}| + (3 + 2\rho)|V_{\text{par}}| + 3$. This means the total number of calls is linear in the size of the graph. We can call literature searches for leaves in parallel (Algorithm 1). In Appendix B, we provide an example run of the logged experiment, which has 6 leaves and 4 parent nodes. The system uses about 40 model calls, which is much more than the 1 or a few calls needed for the standard one-pass report.

For evaluations we use GPT-4.1-mini for all retrieval-based answers and planning, GPT-5 for DAG generation, synthesis, and report writing, and `text-embedding-3-large` for retrieval. All models are accessed through Azure OpenAI. By default, we set a maximum DAG depth of 2 and a maximum number of 50 nodes. To keep the DAG sparse, we also limit the number of children per node to 10. We use 1 retriever call per literature search and 1 refinement round per parent by default.

5 Experiments

We evaluate De(ep)Composition and compare it to OpenAI Deep Research (OpenAI, 2025) and STORM (Shao et al., 2024) using two evaluation methods: (1) an automated debate, where an LLM scores a debate between De(ep)Composition and Deep Research, and (2) a blind human evaluation of final reports from all three systems. The evaluations are small in scale as detailed in Section 9.

We select three debate motions that require different types of reasoning: moral (a moral debate on whether moral systems rooted in theism are preferable to those not rooted in theism), economic and technical (an economic and technical debate on whether the CHIPS and Science Act will enable U.S. semiconductor manufacturers to outperform global competitors over the next decade), and conceptual (a conceptual debate on whether artificial intelligence can be creative in the way humans are). Each debate is run twice, with each system (De(ep)Composition and Deep Research) alternating between the Pro and Con sides in each debate. Thus, we have a total of 6 transcripts and control for position bias.

Metric	De(ep)Composition	OpenAI Deep Research	Difference
Stance consistency (%)	91	68	+23
Logical coherence (1–10)	8.8	7.2	+1.6
Evidence integration (1–10)	8.9	7.1	+1.8
Rebuttal specificity (1–10)	8.7	7.0	+1.7
Persuasiveness (1–10)	8.5	7.6	+0.9
Fluency (1–10)	7.7	8.6	−0.9

Table 1: Automated debate scores, averaged over six transcripts corresponding to three motions and two sides per motion, assessed by GPT-4o-mini.

Metric	De(ep)Composition	OpenAI Deep Research	STORM
Supported-claims rate	0.80	0.68	0.60
Evidence utilization	0.76	0.67	0.61
Logical connection quality (1–10)	7.7	6.5	6.0
Firm stance (of 10 reports)	9	1	2

Table 2: Human assessment scores for ten reports per system, coded from a single blinded human rater. Values are means over reports.

To conduct each debate, each system independently researches its assigned motion. De(ep)Composition stages its arguments based on the nodes of the DAG it builds, while Deep Research uses its standard retrieval and synthesis pipeline. We use GPT-5-mini to stage the debate. Given the research produced by the two systems for each motion, GPT-5-mini debates back and forth as each side would. The debate consists of alternating sides, with 200–300 words each turn, based on the side's own research and the opponent's last turn. GPT-4o-mini scores the debate transcripts on stance consistency (the fraction of turns that maintain the assigned side), logical coherence, integration of evidence, specificity of rebuttal, persuasiveness, and fluency. Each of the other categories is scored on a 1–10 scale, following DeepResearch Bench (unpublished project by Abhinav Chinta, Charles Ding).

Reports on ten motions were assessed by a human reader. These were chosen because they corresponded to motions in which a synthesis of evidence was necessary, and because the report's author had to take a position, rather than simply state facts. The motions concerned ethics, economics, policy, and technology, and are listed in Table 7. Each of the three AI systems wrote a report on each motion, meaning there were 30 reports in total. All of the reports were scored on the same rubric by one of the authors, who was blind to which system wrote each report. The rubric was composed of four measures: the fraction of claims that carry a citation, which we refer to as the supported-claims rate; the fraction of cited sources actually utilized in the argument; the quality of logical connection, on a scale from 1 to 10; and whether the report maintains a firm position throughout.

The rubric also included scores for evidence relevance and originality, but these were at the same level for each of the three systems, and can not be reconstructed. Therefore, these measures were excluded from the analysis. Furthermore, there was only one rater, so score agreement statistics are not applicable. The scores are means over ten reports, and confidence intervals are not reported.

The repository includes all the De(ep)Composition and STORM reports on the ten motions, plus the prompts, pipeline, and single logged run. Deep Research reports and transcripts were not retained. Therefore, these results and discussions in Sections 7 and 8 concern the twenty reports and logged run that were retained.

6 Results

Table 1 shows the automated debate scores. De(ep)Composition scores higher than Deep Research on stance consistency, logical coherence, evidence integration, rebuttal specificity, and persuasiveness. Deep Research scores higher on fluency. The transcripts indicate a

more coherent pattern across the three motions. For theism, De(ep)Composition's Pro case starts from the premise that moral objectivity requires a transcendent anchor, which is justified through sub-claims supported by evidence such as differences in volunteering rates among religious groups. By contrast, Deep Research's Con case invokes the Euthyphro dilemma and repeats it with little additional evidence.

For the reversed sides, De(ep)Composition's Con case appeals to meta-ethical consistency, cross-cultural moral variation, and secular moral psychology. Deep Research's Pro arguments simply reiterate previous points. On the semiconductor motion, De(ep)Composition's Pro case divides the case into quantifiable claims about capital costs with subsidies, integration of research and fabrication, supply chain resilience, and expected yields. The Con case cites quantified dependencies, such as a plateau in the U.S. share of fabrication capacity around 14% by 2032. Deep Research's arguments for both sides are qualitative.

For creativity, De(ep)Composition separates empirical sub-claims comparing divergent thinking benchmarks from the conceptual claim that creativity is novelty plus usefulness. Deep Research argues from metaphysical considerations without an operational definition. Across all six transcripts, De(ep)Composition produces roughly twice as many sub-claims connected to evidence. All of these differences are a consequence of the pipeline structure, which requires every sub-claim to be answered and supported before the case is composed. The only measure that Deep Research scores higher than De(ep)Composition is the fluency score. Deep Research's arguments read more smoothly because De(ep)Composition's heavy reliance on sub-claim citations disrupts the prose flow.

Table 2 shows the human assessment scores. De(ep)Composition has the highest supported-claims rate (0.80 vs. 0.68 for Deep Research and 0.60 for STORM) and the highest fraction of cited sources utilized (0.76 vs. 0.67 for Deep Research and 0.61 for STORM). It also has the highest average score for logical connection (7.7 vs. 6.5 for Deep Research and 6.0 for STORM). De(ep)Composition maintains a firm position in 9 out of 10 reports, compared to 1 report for Deep Research and 2 reports for STORM. The human rater's written comments provide concrete examples of these scoring differences. For the motion on a moral obligation to strangers, De(ep)Composition constructs a multi-layered argument connecting empathy to social trust. Deep Research provides a balanced account that fails to reach a conclusion. STORM outlines positions without establishing causal links.

For the Federal Reserve motion, De(ep)Composition builds a chain from inflation control to wage stability, whereas Deep Research mentions both effects without connecting them. For the foundation model investment motion, De(ep)Composition's argument is conditional: foundation models concentrate capital efficiency, unless saturation dampens marginal returns. Both baseline models lack this conditional structure.

We have plausible explanations for the two differences. More reported claims are supported because every leaf answer has its own citations, while all parent contributions are derived from its children, giving little opportunity for uncited claims. Note that the measure only checks whether a claim has a citation in it, not whether the citation is accurate, which we discuss in Section 8. Similarly, a higher proportion of firm positions partly reflects the report prompt, which instructs the model to defend the root answer (we return to this in Section 9).

7 Analysis of the Reports

Table 2 reports scores from a single rater. These ratings are thus subjective and tied to that rater's judgment. However, there are twenty reports on ten motions under consideration (ten De(ep)Composition reports and ten STORM reports on the same motions), allowing us to assess them without a rater. The ten De(ep)Composition and ten STORM reports are measured using the same procedures on each motion, and comparing them motion-by-motion controls for motion difficulty. We did not keep the Deep Research set, so we only compare De(ep)Composition and STORM here. All reported differences are medians of motion-wise differences, with 95% bootstrap intervals (Efron & Tibshirani, 1993) and

Measure	De(ep)Composition	STORM	Difference	95% CI	Wins
Citations per 100 words	5.7	4.7	+1.0	[+0.0, +2.1]	8/10
Sentences with a citation	0.46	0.60	−0.15	[−0.23, −0.09]	0/10
Distinct sources	8	27	−18	[−24, −14]	0/10
Citations per source	9.3	4.1	+5.0	[+1.7, +7.2]	10/10
Hedges per 1,000 words	2.8	6.4	−2.2	[−7.3, −0.4]	1/10
Contrast markers per 1,000 words	5.5	6.0	−0.3	[−0.8, +1.0]	4/10
Commitment markers per 1,000 words	6.8	3.8	+3.3	[+0.4, +4.2]	10/10
Causal connectives per 1,000 words	4.2	3.0	+0.3	[−0.5, +1.8]	5/10
Words	1339	2426	−1172	[−1621, −617]	0/10

Table 3: De(ep)Composition vs. STORM paired comparison. Values are medians over motions, computed from the ten De(ep)Composition reports and ten STORM reports. Reported difference is the median over motions of motion-wise differences (along with a 95% bootstrap interval). "Wins" indicates how many motions for which the De(ep)Composition score is higher. The reported rates are based on the word lists in Appendix F.

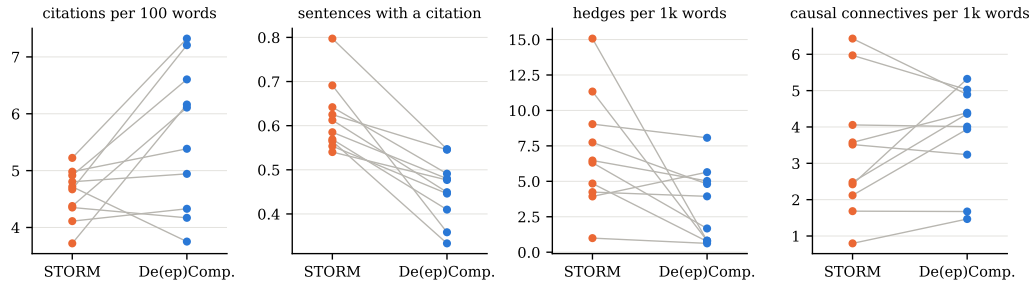


Figure 2: 4 measures per motion, linked across STORM and De(ep)Composition, left to right.

Wilcoxon signed-rank tests (Wilcoxon, 1945). With only ten motion pairs, the smallest attainable two-sided p-value is 0.002. (We strip bibliographies, headings, and markup before counting.)

Assuming there is a detectable stance, we would expect it to manifest in the language. Hedges like *may*, *could*, and *suggests* withhold commitment (Hyland, 2005), whereas markers like *must*, *demonstrates*, and *therefore* assert it. The De(ep)Composition reports use fewer hedges (2.8 per 1,000 words versus STORM's 6.4 per 1,000 words; De(ep)Composition uses fewer on nine of the ten motions; $p = 0.020$) and more commitment markers (6.8 versus 3.8 per 1,000 words; De(ep)Composition uses more on all ten motions; $p = 0.002$).

Although contrast markers like *however* and *on the other hand* are roughly equivalent (5.5 versus 6.0 per 1,000 words; $p = 0.922$), so that De(ep)Composition is just as likely to acknowledge the other side, the strength of commitment differs.

Commitment is strongest at the start and weakest at the end of a report. Table 5 provides the side chosen by each De(ep)Composition report, whether the report title makes a claim, and the nature of the conclusion in each report's final paragraph. Nine of ten titles make a claim, as instructed by the prompt. Each of the STORM reports begins with a lead section presenting the question as a debate. The final paragraphs are notably less committed: five firm, two qualifying the claim in the title, two conditionally qualifying it, and one a compromise instead of a yes/no response.

The creativity report concludes that the irreducibility of human creativity persists, "but only provisionally." The theism report concludes that "the preference for theistic morality depends fundamentally on one's valuation of transcendent moral truth versus pragmatic ethical function." One report actually argues against the motion, that application-layer startups will outperform foundation model developers. A plausible explanation is that,

Source type	Sources	Share
Government	45	13%
Academic	74	21%
Think tank or NGO	78	22%
News or magazine	3	1%
Encyclopedia	4	1%
Company, blog, or other	145	42%

Table 4: The domains of the 349 sources in ten De(ep)Composition bibliographies, classified by host name.

although the stance instruction does guide the title and executive summary (both written first), the conclusion is written last from a graph whose branches explore both sides of the question.

The rater marked De(ep)Composition higher on logical connection and attributed this to causal connections between sub-claims. We can use causal connectives like *because*, *leads to*, and *as a result* as a rough proxy. The rate is not reliably different (De(ep)Composition, 4.2 per 1,000 words; STORM, 3.0 per 1,000 words; De(ep)Composition uses more on five motions; $p = 0.432$). Note that this captures only a part of logical connection and coherence; we neither refute the rater nor provide independent evidence supporting their claims about a causal chain.

De(ep)Composition reports are more compact than those of STORM: its median report length is 1339 words rather than 2426. It cites more frequently: at 5.7 citations per 100 words, higher than STORM's 4.7 ($p = 0.037$). However, De(ep)Composition draws on a more restricted range of sources: each De(ep)Composition report cites a median of just 8 sources, compared with 27 for STORM. On average, De(ep)Composition cites each source 9.3 times per report, whereas STORM cites each source 4.1 times. In De(ep)Composition, only 0.46 of sentences in each report have citations (compared with 0.60 in STORM).

Rather, the evidence citations are concentrated in bullet points, while the narrative linking those points consists of uncited text. We can see from the bibliographies of De(ep)Composition reports that a large number of retrieved sources will not be cited in the report. The median De(ep)Composition bibliography contains 36 sources (range 30–39), of which just 7–13 are cited (a median of 25%).

Table 4 provides a domain breakdown of the 349 sources, showing that 13% are from government sites, 21% from academic (university) sites, 22% from think tanks and non-profits and 42% from company sites, blogs and other. There are almost no news sites. We did a test for link validity in October 2026, 11 months after the reports were written: 69% of links resolved correctly, 27% refused automation, 4% had become extinct or returned error. We did not check on STORM's export reports, as they do not include URLs.

In terms of report structure, STORM reports follow the structure of an encyclopedia article: with a short lead section; a background section; sections on arguments for and/or against the motion, often ending with the opposing case. De(ep)Composition reports are written in the form of a brief, with the claim as report title, a short executive summary containing the answer, sections based on branches of the graph, ending with a conclusion. For example, on African Continental Free Trade Area, STORM leads with a description of the agreement and its ratification history, whereas De(ep)Composition leads with the conclusion that "evidence overwhelmingly supports that its benefits outweigh the harms."

For the motion that individuals have a moral duty to treat strangers equally to people they know, STORM ends with a description of critics deeming an equal obligation to strangers unrealistic, whereas De(ep)Composition ends with a synthesis of "impartiality and partiality are not opposites", which is more moderate than the report title suggests.

Motion	Side	Title	Final paragraph
Human creativity will not remain fundamentally irreducible to …	for	claim	qualified
Individuals have a moral duty to prioritize the well-being of …	for	claim	qualified
Investing in companies developing foundation models -e.g., OpenAI, …	against	claim	firm
Israel and Palestine should pursue a long-term joint economic zone …	for	claim	firm
It is ethically justified for hedge funds to use alternative data …	for	topic	conditional
Moral systems rooted in theism are preferable to non-theistic …	for	claim	conditional
The benefits of the African Continental Free Trade Area outweigh …	for	claim	firm
The U.S. Federal Reserve should prioritize maintaining high …	neither	claim	compromise
The U.S. should develop wind, solar, and geothermal energy …	for	claim	firm
U.S. semiconductor manufacturers will outperform global peers over …	for	claim	firm

Table 5: Position of the reported De(ep)Composition reports. We specify the position as it is relative to the wording of the motion: *firm* reports repeat the claims in the motion's title; *qualified* reports restrict these claims; *conditional* reports make them dependent on specific conditions or values that are not part of the motion, and *compromise* reports take a position on neither side.

8 Citation Fidelity

The rubric used to grade De(ep)Composition reports in Section 5 scored claims with citations as supported, but did not check that the cited source actually made the claim. Liu et al. (2023) found that around one quarter of the citations produced by generative search engines did not support their sentences. This section measures the citation fidelity of De(ep)Composition reports. It first introduces an automatic test for checking citation fidelity, validates it on correctly-cited claims, and then applies it to all De(ep)Composition reports to find where the link to sources is lost.

8.1 A lexical attribution test

The test exploits the intuition that a particular sentence in a report is likely to use some unusual words from its source. The test calculates for every sentence in a report that has a citation, the TF-IDF cosine similarity (Salton & Buckley, 1988) of that sentence with each source in the document, and ranks the sources. It then assigns a percentile rank to each cited source based on its ranking (0 is the best match, 1 is the worst match).

If the citations are correct, the cited sources would tend to be highly ranked (low percentile) whereas if they were random, the percentile rank would be distributed uniformly between 0 and 1 (with median 0.5). We report the results as the median percentile rank of all cited sources and as the proportion of claims where the single best matching source is cited, compared to the chance expectation. The test is model-free and can run on all reports. Being lexical, the test risks under-ranking a source if it uses different wording, a practical limitation we measure in the validation.

A sensible validation is to ask whether the test can identify citations known to be correct. The leaf answers in the running log (Appendix B) provide a ground truth by construction: each was generated by a single model call and the excerpts it cited were retrieved directly from the relevant documents. Against the set of all 26 sources used by the run, the cited source is the best match for 96% of leaf claims. (Note that a ratio of 10% is expected by

Setting	Claim–source pairs	Median rank percentile	Top-3 hit rate	Chance	Best match cited
Leaf answers, own numbering	63	0.04 [0.00, 0.08]	0.83	0.12	0.96 (chance 0.10)
Leaf answers, own numbering, full pages	24	0.00 [0.00, 0.11]	0.88	0.30	0.84 (chance 0.13)
Leaf answers as a parent reads them, global numbering	63	0.04 [0.00, 0.08]	0.83	0.12	0.96 (chance 0.10)
Leaf answers as a parent read them, original pipeline	63	0.12 [0.08, 0.16]	0.49	0.12	0.52 (chance 0.10)
Final report, logged run	173	0.48 [0.44, 0.52]	0.06	0.12	0.04 (chance 0.10)
Evaluated reports, all listed sources	491	0.47 [0.40, 0.53]	0.18	0.14	0.14 (chance 0.10)
Evaluated reports, cited sources only	491	0.43 [0.40, 0.50]	0.56	0.51	0.41 (chance 0.36)

Table 6: Attributions of cited claims. Each row ranks all available sources according to the lexical similarity between their text and the text of each cited claim. The cited sources have median rank percentile 0 if always cited correctly, and 0.5 if citations are random (95% bootstrap confidence interval in brackets). Top-3 hit rate is the fraction of cited sources that rank in the top 3. The column of chance values shows the top-3 hit rate expected under random citation. The final column shows the fraction of claims for which the best match was cited.

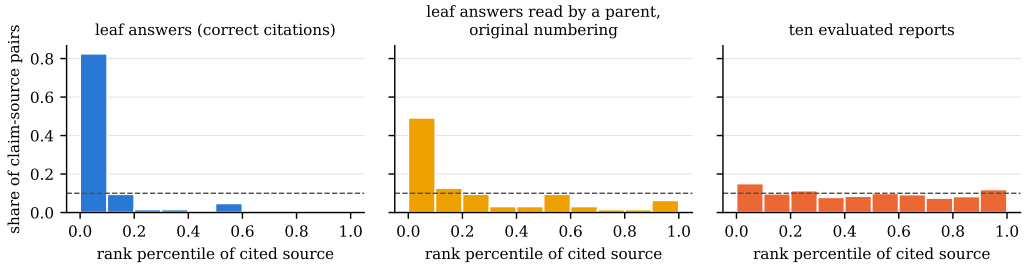


Figure 3: Rank percentiles of cited sources. If citations are correct, then they cluster around 0 on the left-hand side of the figure. In contrast, the distributions of percentiles of the evaluated reports' citations are flat as expected by random citation on the right-hand side. The middle panel shows the percentiles of the leaf answers as read by a parent node during composition, Section 8.3. A dashed line indicates the uniform distribution.

chance.) The median percentile of the cited sources is 4% (95% confidence interval: [0%, 8%]).

Our evaluated reports cite full web pages, which means that a web page cited as a source also contains text unrelated to the claim. We repeated the validation test with the full text of each page. Against the 10 pages containing usable text, the best match is cited as source for 84% of claims, 13% by chance. The test again distinguishes between correct citations and random ones.

8.2 Results

Table 6 and Figure 3 summarise the results. Our 10 evaluated reports contain 235 citations of claims and 491 (source, claim)-pairs. Against each report's listed sources, the median percentile of cited sources is 0.47 (0.40 to 0.53), indistinguishable from the 0.5 expected under random citation. Against that same set, the best match is cited as source for 14% of claims, by chance 10%. If we limit our comparison to the sources each report actually cites (a median of 6 per report), the picture does not change: the best match is cited as source for 41% of claims, by chance 36%.

Finally, the last report produced by our logged run, which was produced using the same report-writing step, continues to perform at chance, with a median percentile of 0.48. The information provided by the evaluated reports' citations is, in terms of our validation test, no more than random guesses about the source supporting each claim.

For instance, a bullet point stating that "non-theistic systems such as Confucianism, Buddhism, and Stoicism achieved moral stability" without divine law cites a collection of 5

sources that argue from the moral law for the existence of God. None of these sources are the origin of the claim. Similarly, a bullet point on ancient Egyptian, Israelite, and Vedic societies cites a work titled "A Beginner's Guide to the Moral Argument". While the claims may be correct, the citations are not.

8.3 Mechanism

The code for the evaluated reports is not available (Appendix C). However, a snapshot of the repository, taken a week earlier, comes with a run that we have used above. We find that this snapshot of the system loses the link between claims and sources at two steps, both of which are consistent with the pattern we observed in the evaluated reports, though we cannot confirm them on the code we have for the evaluated system.

Local numbers read against a merged list. Each leaf answer numbers its own sources locally, as [1] through to [k]. When the parent of the leaf is composed, the pipeline merges the sources of its children into a single, deduplicated list, numbered [1] through to [m]. Then the synthesis model is instructed to cite these numbers from the merged list. When the children's answers are passed to the parent node, they retain their original local numbers. Thus, when the second child's answer contains "[1]" and that is copied into the parent, it is the first source from the first child in the merged list, not the first source from the second child. The central rows of Table 6 demonstrate this step on our logged run.

Against the parent's merged list, the best match is cited for only 52% of leaf claims, mostly those taken from the first child of each parent. When we restore the leaf answers' local numbering, the 52% jumps back to 96%. Since each level in the tree repeats this merge, the misattribution grows as answers propagate up the graph towards the root.

Node identifiers in place of sources. Note that the composition instructions for Phase 1 required each child to be cited by node identifier, so that downstream steps could track their dependencies. But the synthesis prompt asks the model to cite sources by number, which is how citations appear in the final report. (This is another conflict in instructions that should be addressed.) In the logged run, in which 4 nodes were composed, including the root, three of them, including the root, followed the composition instructions, writing source markers as "(node_3)" with no source number. As a result, the report writer received a root answer without citations but with a command to number them, and it then assigned them arbitrarily. This explains why the final report in the logged run performed no better than chance.

8.4 Correction

This problem and the other one described in the preceding section can be solved by assigning a single, persistent identity to each source throughout the course of the run. In the corrected pipeline, each source is assigned a single global number the first time it is retrieved by any node. Leaf answers are translated from local identifiers to global numbers before being stored, so that all nodes in the run share the same numbering and the report's bibliography does too. After each composition, node identifiers are translated to global citations for the children in the parent answer, and any numbers not associated with the parent's sources are dropped. The synthesis and composition-instruction prompts are then modified to require citation by source number rather than node identifier.

If we replay the logged run with this correction, leaf claims are cited by their best matching source 96% of the time, just as in the leaf itself. (This study does not have any way to determine whether the attribution is preserved end to end through the report-writing step, and new experiments would be required to establish that, but the current tests make such a check inexpensive.)

8.5 Implications

Finally, the rate of supported claims in Table 2 is computed by counting citation markers. This metric measures the frequency with which De(ep)Composition adds citations, which it does more densely than STORM (Section 7), but it says nothing about whether the claims are actually supported by the cited sources. We should not assume that the reports are more "grounded" than the baselines on this basis. This test is easy to apply to any system that produces inline citations, but we were not able to apply it to either baseline, because the exports from STORM did not include URLs and Deep Research reports were not retained.

9 Discussion

The results presented here strongly suggest three things about De(ep)Composition. First, it is able to write decisive reports. The rater found a firm position more often for De(ep)Composition than for either baseline, and the lexical data also show fewer hedges on nine out of ten motions and more markers of commitment on all ten motions. Second, De(ep)Composition is able to write compact, claim-centered reports, with citations densely interspersed. Third, the recursive composition in the version of De(ep)Composition evaluated in this paper is not able to maintain the connection between a claim and its supporting source. The higher rate of supported claims and evidence use found by the rater is due to the method's tendency to include consistent citations in a compact report, not to correct attribution of claims to sources.

The firm stance of De(ep)Composition reports is partly an artifact of the prompt. This prompt asks the model to argue for the root answer and avoid hedging, which would be expected to lead to many firm positions, even without the decomposition. But the decomposition provides a definite-format root answer for the prompt, and the experiments described here are insufficient to isolate its effect. (Section 7 also noted that the titles and executive summaries are indeed committed, while half of the conclusions qualify them.) But a likely cause is that the final writing draws on a graph covering both sides of the question, so its balanced content reappears.

To isolate these effects, it would be useful to run three ablations: one that keeps the graph but removes the instruction to take a stance, one that keeps the instruction but removes the graph, and one that requires the conclusion to be in agreement with the title.

The failure of provenance in stage-based composition is due to the fact that a citation number is a pointer to a location in a list of sources. At each level of the graph in the pipeline evaluated in this paper, the source list was being regenerated. It rebuilt the list, but copied the pointers without changing them, so that each pointer in the next stage referenced a different source than the one it had originally pointed to. This is a general risk for any system that has multiple stages of composition with text containing inline citations. For example, a summarizer of summaries or a merging agent integrating outputs from several other agents could make this mistake. And it is invisible: the resulting report will still be well cited, and no reader or automated rubric counting citation markers would notice.

This is avoided by applying two design rules: (1) each source should have a single identifier throughout the course of the entire run and (2) no model should refer to an intermediate result by an identifier that cannot ultimately be resolved to a source. It is easy to verify that the rules are followed using a lightweight lexical audit, such as the one described in Section 8. (Attribution metrics based on entailment models (Rashkin et al., 2023, Gao et al., 2023) would provide a stronger check.)

The study uses ten motions and six transcripts, which is too small to make accurate estimates. The human scores are given by a single human rater who is blind to the system but is a coauthor and knows about the study. The model-based debate evaluation uses a small model with the unpublished course rubric, and two measures of the human rubric were discarded because their data records were unusable. The paired measurements discussed in Section 7 are not dependent on human raters, but are lexical proxies. The stance classifications in Table 5 were produced by a single evaluation.

The three systems use different models, search backends, and prompts, so the comparison is a systems one and not just a decomposition one. STORM was designed to write balanced encyclopedia articles, so its poor performance on firm positions on this task is not a flaw. Deep Research's output is in a commercial tool, so its internal model and version are uncontrolled. A controlled test would fix the language model and search API on a flat search agent, STORM, and De(ep)Composition; include the ablations cited here; and measure attribution as in Section 8 for all systems alongside human judgment.

A De(ep)Composition run requires roughly four model calls per leaf and a few per parent node (Section 4), so its total cost increases linearly with the graph size and is many times higher than a one-shot report. Most of the retrieved sources are not used; only 25% of the listed sources are in the body of the report, on median. A bibliography can be shortened without affecting the argument by excluding sources that are not used by any claim.

10 Conclusion

De(ep)Composition reduces a question to a sub-question graph and answers it bottom-up under generated composition instructions, grows the graph when answers are not complete, and finally argues the result in a report. It provided more decisive, densely cited arguments than Deep Research and STORM on ten contested motions and six automated debates, but fluency decreased and conclusions were often more qualified than their titles suggested. An automatic attribution test showed that the citations in its evaluated reports do not point to the sources of the claims, since the citation numbers become unintelligible as answers are composed.

Assigning each source a single identifier for the entire run would avoid this failure mode. A controlled comparison on the same language model and search API with the ablations mentioned here, and with more motions and raters, and measuring attribution for all systems is the next step.

References

Maciej Besta, Nils Blach, Ales Kubicek, Robert Gerstenberger, Michal Podstawski, Lukas Gianinazzi, Joanna Gajda, Tomasz Lehmann, Hubert Niewiadomski, Piotr Nyczyk, and Torsten Hoefler. Graph of thoughts: Solving elaborate problems with large language models. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 38, pp. 17682–17690, 2024.

Bradley Efron and Robert J. Tibshirani. *An Introduction to the Bootstrap*. Chapman and Hall/CRC, 1993.

Tianyu Gao, Howard Yen, Jiatong Yu, and Danqi Chen. Enabling large language models to generate text with citations. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pp. 6465–6488, 2023.

Xinyu Hua and Lu Wang. Neural argument generation augmented with externally retrieved evidence. In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 219–230, 2018. URL https://aclanthology.org/P18-1021.

Ken Hyland. *Metadiscourse: Exploring Interaction in Writing*. Continuum, London, 2005.

Omar Khattab, Arnav Singhvi, Paridhi Maheshwari, Zhiyuan Zhang, Keshav Santhanam, Sri Vardhamanan, Saiful Haq, Ashutosh Sharma, Thomas T. Joshi, Hanna Moazam, Heather Miller, Matei Zaharia, and Christopher Potts. DSPy: Compiling declarative language model calls into state-of-the-art pipelines. In *International Conference on Learning Representations*, 2024.

Tushar Khot, Harsh Trivedi, Matthew Finlayson, Yao Fu, Kyle Richardson, Peter Clark, and Ashish Sabharwal. Decomposed prompting: A modular approach for solving complex tasks. In *International Conference on Learning Representations*, 2023.

John Lawrence and Chris Reed. Argument mining: A survey. *Computational Linguistics*, 45 (4):765–818, 2020. URL https://aclanthology.org/J19-4006.

Guohao Li, Hasan Abed Al Kader Hammoud, Hani Itani, Dmitrii Khizbullin, and Bernard Ghanem. CAMEL: Communicative agents for "mind" exploration of large language model society. In *Advances in Neural Information Processing Systems*, volume 36, 2023.

Nelson F. Liu, Tianyi Zhang, and Percy Liang. Evaluating verifiability in generative search engines. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pp. 7001–7025, 2023.

OpenAI. Introducing deep research. https://openai.com/index/introducing-deep-research/, February 2025.

Ofir Press, Muru Zhang, Sewon Min, Ludwig Schmidt, Noah A. Smith, and Mike Lewis. Measuring and narrowing the compositionality gap in language models. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pp. 5687–5711, 2023.

Hannah Rashkin, Vitaly Nikolaev, Matthew Lamm, Lora Aroyo, Michael Collins, Dipanjan Das, Slav Petrov, Gaurav Singh Tomar, Iulia Turc, and David Reitter. Measuring attribution in natural language generation models. *Computational Linguistics*, 49(4):777–840, 2023.

Gerard Salton and Christopher Buckley. Term-weighting approaches in automatic text retrieval. *Information Processing & Management*, 24(5):513–523, 1988.

Yijia Shao, Yucheng Jiang, Theodore A. Kanell, Peter Xu, Omar Khattab, and Monica S. Lam. Assisting in writing Wikipedia-like articles from scratch with large language models. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pp. 6252–6278, 2024. URL https://aclanthology.org/2024.naacl-long.347.

Noam Slonim, Yonatan Bilu, Carlos Alzate, Roy Bar-Haim, Ben Bogin, Francesca Bonin, Leshem Choshen, Edo Cohen-Karlik, Lena Dankin, Lilach Edelstein, et al. An autonomous debating system. *Nature*, 591(7850):379–384, 2021. doi: 10.1038/s41586-021-03215-w.

Jason Wei, Zhiqing Sun, Spencer Papay, Scott McKinney, Jeffrey Han, Isa Fulford, Hyung Won Chung, Alex Tachard Passos, William Fedus, and Amelia Glaese. BrowseComp: A simple yet challenging benchmark for browsing agents. *arXiv preprint arXiv:2504.12516*, 2025.

Frank Wilcoxon. Individual comparisons by ranking methods. *Biometrics Bulletin*, 1(6): 80–83, 1945.

Qingyun Wu, Gagan Bansal, Jieyu Zhang, Yiran Wu, Beibin Li, Erkang Zhu, Li Jiang, Xiaoyun Zhang, Shaokun Zhang, Jiale Liu, Ahmed Hassan Awadallah, Ryen W. White, Doug Burger, and Chi Wang. AutoGen: Enabling next-gen LLM applications via multi-agent conversation. *arXiv preprint arXiv:2308.08155*, 2023.

Shunyu Yao, Dian Yu, Jeffrey Zhao, Izhak Shafran, Thomas L. Griffiths, Yuan Cao, and Karthik Narasimhan. Tree of thoughts: Deliberate problem solving with large language models. In *Advances in Neural Information Processing Systems*, volume 36, 2023a.

Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. ReAct: Synergizing reasoning and acting in language models. In *International Conference on Learning Representations*, 2023b.

Denny Zhou, Nathanael Schärli, Le Hou, Jason Wei, Nathan Scales, Xuezhi Wang, Dale Schuurmans, Claire Cui, Olivier Bousquet, Quoc V. Le, and Ed H. Chi. Least-to-most prompting enables complex reasoning in large language models. In *International Conference on Learning Representations*, 2023.

Mingchen Zhuge, Haozhe Liu, Francesco Faccio, Dylan R. Ashley, Róbert Csordás, Anand Gopalakrishnan, Abdullah Hamdi, Hasan Abed Al Kader Hammoud, Vincent Herrmann, Kazuki Irie, Louis Kirsch, Bing Li, Guohao Li, Shuming Liu, Jinjie Mai, Piotr Piękos, Aditya Ramesh, Imanol Schlag, Weimin Shi, Aleksandar Stanić, Wenyi Wang, Yuhui Wang, Mengmeng Xu, Deng-Ping Fan, Bernard Ghanem, and Jürgen Schmidhuber. Mindstorms in natural language-based societies of mind. *arXiv preprint arXiv:2305.17066*, 2023.

A Prompts

We reproduce the instructions used for the DSPy signatures on the evaluation runs, without the input and output fields, and concatenating multi-line lists into sentences.

LayerChildGenerationSignature (Phase 1, one layer of children)

```
Generate children for a specific parent node in the research DAG.
You are a research planning expert. Given a parent research question and the current
    DAG structure, generate focused sub-questions that can be answered independently
    to collectively answer the parent question. Look at the entire DAG context to
    ensure coherence and avoid duplication.
Guidelines:
1. Generate focused, standalone sub-questions that collectively answer the parent
2. Consider the existing DAG structure to maintain coherence
3. Respect the max_children limit strictly
4. Each child should be answerable independently
5. Output format should always be "list" (bullet points)
```

CompositionInstructionSignature (Phase 1, per parent)

```
Write explicit parent composition instructions referencing each child node.
Given a parent node and its children, explain exactly how the child answers will be
    combined to produce the parent's output. Always reference children by their IDs
    so downstream steps can trace dependencies unambiguously.
```

LeafNodeResearcher (Phase 2, leaves)

```
Answer a leaf node research task using literature search results.
Given literature search results and the expected output format, provide an answer
    that strictly adheres to the format requirements.
Format types and requirements:
- boolean: Provide "Yes." or "No." followed by a single sentence justification with
    citations.
- short_answer/list/report: Provide 3-5 bullet points ('- ' prefix) summarizing key
    takeaways with citations.
- table_csv/json: Provide structured data (CSV or JSON) with inline citations in
    headers or cells.
CRITICAL: Preserve all citations from the literature search in [X] format. Do not add
    citations that aren't in the source material.
```

ParentNodeSynthesizer (Phase 2, parents)

```
Synthesize answers from child nodes to answer a parent node's research task.
Given the answers from all child subtasks and composition instructions, combine them
    to produce the answer for the parent task in the required format.
CRITICAL RULES:
1. Use ONLY the information from the child results - no external knowledge
2. Follow the composition instructions exactly
3. When citing evidence, use the citation numbers [1], [2], etc. from the AVAILABLE
    CITATIONS section - do NOT reference child nodes by name
4. Output in the specified format
5. If child results contradict, acknowledge and explain the contradiction
```

GapExplorer (Phase 2, refinement, first pass)

```
Explore potential gaps that could enhance the composed answer.
This is the FIRST PASS - be creative and identify any areas where the answer could be
    improved, even if they're minor enhancements. Don't filter yet.
Think about: missing perspectives or viewpoints. Unaddressed sub-questions or
    aspects. Areas that could use more depth or detail. Counterarguments or
    alternative viewpoints not covered. Quantitative data or evidence gaps. Temporal
    or geographic coverage gaps. Edge cases or exceptions not mentioned.
```

```
Return a list of potential gaps, even if they seem minor. The second pass will
    determine which ones are critical enough to warrant refinement.
```

GapImportanceEvaluator (Phase 2, refinement, second pass)

```
Evaluate whether identified gaps are important enough to warrant refinement.
This is the SECOND PASS - evaluate gaps based on the specified conservatism level.
CONSERVATISM MODES:
- "conservative": Only flag gaps that are CRITICAL - the answer would be misleading
    or fundamentally incomplete without this information. Default to NOT refining.
- "balanced": Flag gaps that would SIGNIFICANTLY improve the answer's completeness,
    depth, or accuracy. Be moderate - refine if the gap meaningfully enhances quality.
- "aggressive": Flag gaps that would improve the answer, even if moderately. Refine
    if the gap would add value, even if not critical.
Consider: Would adding this information fundamentally change the answer's conclusion?
    Is this gap misleading or would it cause the answer to be wrong? Would a
    reasonable reader say the answer is incomplete without this? Is this a core
    aspect of the question that's missing? How much would this gap improve the
    answer's quality?
Provide a confidence score (0.0-1.0) indicating how confident you are that refinement
    is warranted.
```

OutlineFromDAGSignature (Phase 3)

```
Generate a report outline that structures a persuasive argument from DAG research.
The DAG represents a research question decomposed into sub-questions. Each node
    answered a question through research. The root node has the final answer. Child
    nodes contain supporting evidence that led to that answer.
Create an outline for a persuasive report arguing a definitive stance for the root
    answer. Use the DAG's hierarchy to organize evidence: child findings support
    parent conclusions. Structure as a logical argument (Executive Summary, Evidence
    Sections, Conclusion).
REQUIREMENTS: Start with Executive Summary stating root answer prominently. Organize
    body sections to build argument logically using DAG hierarchy. Use ## for main
    sections, ### for subsections. Design for prose writing (paragraphs, not just
    lists). Base outline ONLY on content actually present in root_answer (no
    placeholders).
```

ReportFromDAGSignature (Phase 3)

```
Generate a comprehensive final report from the DAG results.
The report must: 1. Start with the root answer prominently (matching its format) 2.
    Follow the provided outline structure 3. Incorporate findings from all DAG nodes
    4. Take a clear stance that matches the root answer 5. Preserve all citations [X]
    from the DAG 6. Be well-written, coherent, and comprehensive
CRITICAL STANCE ALIGNMENT: If root answer is "Yes" (boolean), the report must argue
    FOR that position. If root answer is "No" (boolean), the report must argue
    AGAINST. If root answer is a list, the report should be organized around those
    items. The report's conclusion must align with the root answer. Do not contradict
    or hedge against the root answer's stance.
FORMATTING REQUIREMENTS: The report title (# heading) should be the CLAIM, not the
    question. Use proper markdown. Write in prose paragraphs with supporting bullet
    points where appropriate. Do not include internal node references in the output.
    Only use citation numbers like [1], [2], [3]. Every section must contain
    substantive content from the dag_results. Do not include a bibliography section
    (it is added later).
```

B An Example DAG

Figure 4 shows a DAG from a logged run that considered whether the US federal government should substantially increase social services for persons living in poverty. In this

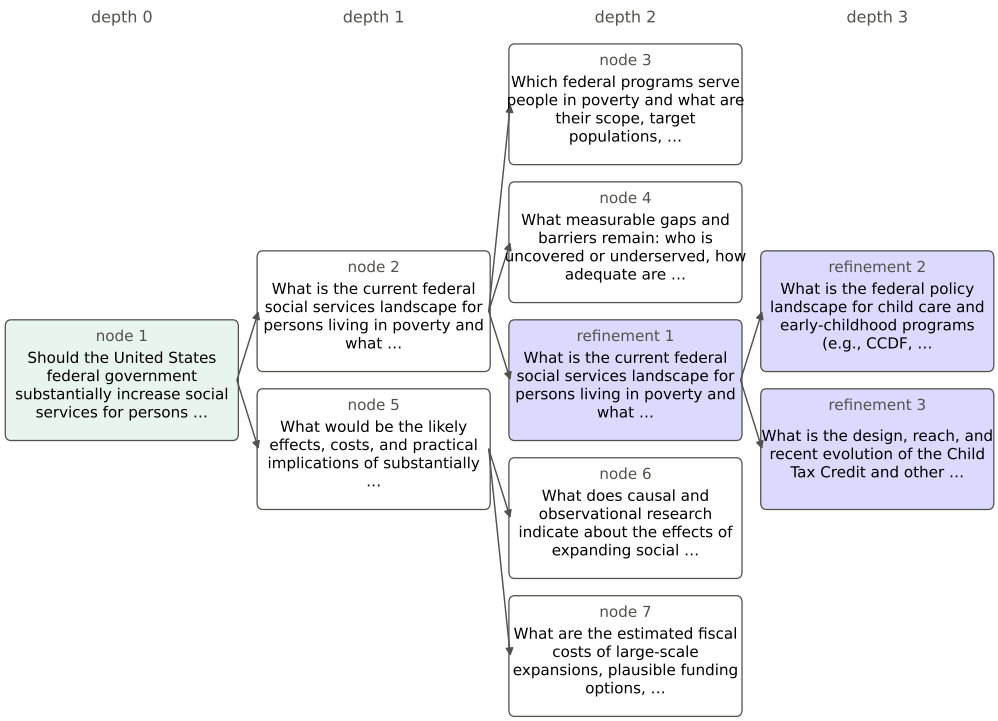


Figure 4: A generated and refined DAG. Arrows indicate parent → child. Nodes shaded were added when the refinement step found the composed answer for node 2 to be incomplete.

example run, we set the DAG depth to 2. The initial planner decomposed the question into the current landscape of federal services, and the likely effects of increasing services. Each of those two topics were further decomposed into 2 leaf questions. After the composition of the landscape node, the refinement check found that child care and family tax credit information was missing, so the pipeline attached a refinement subtree with 2 new leaves (shown shaded).

Across the 6 leaves of this run, we retrieved 26 distinct sources (4.3 per leaf, on average). Of the 26 sources available at the root, 9 of them were from the refinement subtree. Thus, the refinement step contributed about a third of the total evidence base. The final report has 3154 words, 213 citations, and cites all 26 sources. Section 8 shows that the citations do not match the claims they support. We provide the full log in the repository, which shows the answers and citations of each node.

C Code Versions

There are three versions of the pipeline relevant to this study, two of which we have available.

1. **Evaluated version.** The reports we included in Sections 5 and 7 were finalized on 2 December 2025. The prompts for this version are reproduced in Appendix A, and are the same as those included in our course report. In this version of the pipeline, the DAG is built one layer at a time, composition instructions are written on a second pass, gaps are identified by a two-pass explorer and evaluator, and the bibliography separates cited and uncited sources. We do not include the code for this version in the public repository.

Motion	De(ep)Composition			STORM	
	words	cited	listed	words	cited
Human creativity will not remain fundamentally irreducible to artificial intelligence when…	1,235	8	31	1,991	32
Individuals have a moral duty to prioritize the well-being of strangers equally to the…	1,523	8	38	1,887	22
Investing in companies developing foundation models -e.g., OpenAI, Anthropic- will yield…	1,247	9	37	2,712	30
Israel and Palestine should pursue a long-term joint economic zone to promote regional…	1,363	9	35	3,763	35
It is ethically justified for hedge funds to use alternative data sources (e.g., satellite…	1,594	10	30	2,797	24
Moral systems rooted in theism are preferable to non-theistic moral systems.	1,393	11	36	2,010	23
The benefits of the African Continental Free Trade Area outweigh the harms	1,195	7	38	2,376	32
The U.S. Federal Reserve should prioritize maintaining high interest rates over stimulating…	1,226	7	32	3,265	24
The U.S. should develop wind, solar, and geothermal energy projects in the Arctic to…	1,315	13	33	2,477	32
U.S. semiconductor manufacturers will outperform global peers over the next decade due to…	1,605	8	39	2,014	23

Table 7: Word count and number of unique sources cited in the body of each report retained in the analysis. For De(ep)Composition we also include the number of sources listed in the bibliography.

2. **Repository version.** This is the state of the code as of the last commit in the repository (26 November 2025), which generates the entire DAG and its composition directions in a single call (`FullDAGGenerationSignature`). In this version a single validator (`CompositionValidator`) is used to check for gaps, and it reports none unless a core aspect of the question is missing. The logged run in Appendix B was performed with this version, and the citation processing in this version is what we analyze in Section 8.3. The course report mentions that citations were given global numbers later on, but this does not appear to have been done in this version.
3. **Corrected version.** This is the state of the code in the public repository, which is the repository version made configurable for the model provider, with unused code removed, and with the correction described in Section 8. This correction consists of 5 steps: assigning global numbers to citations, replacing node references with source citations, revising prompts, adding an attribution audit (`utils/citations.py`, `utils/attribution_audit.py`), and including tests to replay the logged run.

D Report Statistics by Motion

Table 7 shows the word count and number of unique sources cited in the body of each report retained in the analysis. The medians in Table 3 are based on these metrics.

E Stance Labels

Stance labels in Table 5 are assigned by reading the title and final paragraph of each De(ep)Composition report. We store each label and a short quote of the final paragraph

of each report in the repository (`data/reports/stance_labels.json`), so that all labels can be verified against the report text.

F Word Lists

Rates in Table 3 are based on whole-word, case-insensitive counts per 1,000 body words. The word lists are organized according to the categories in Hyland (2005) as they apply to argumentative reports.

Hedges. may, might, could, possibly, perhaps, arguably, potentially, appears, appear, seems, seem, suggests, suggest, likely, unlikely, uncertain, unclear, debated, contested, controversial, depends, depending, can be, to some extent, in some cases, remains open.

Contrast markers. however, on the other hand, conversely, while, whereas, although, though, nevertheless, nonetheless, critics, opponents, proponents, detractors, both sides.

Commitment markers. must, should, clearly, demonstrates, demonstrate, shows, show, proves, establishes, therefore, thus, decisively, undeniably, strongly, essential, necessary, outweigh, outweighs, superior, preferable.

Causal connectives. because, therefore, thus, hence, as a result, consequently, leads to, lead to, led to, results in, result in, resulting in, drives, drive, causes, cause, due to, which means, so that, enables, enable, reduces, increases, raises, lowers.